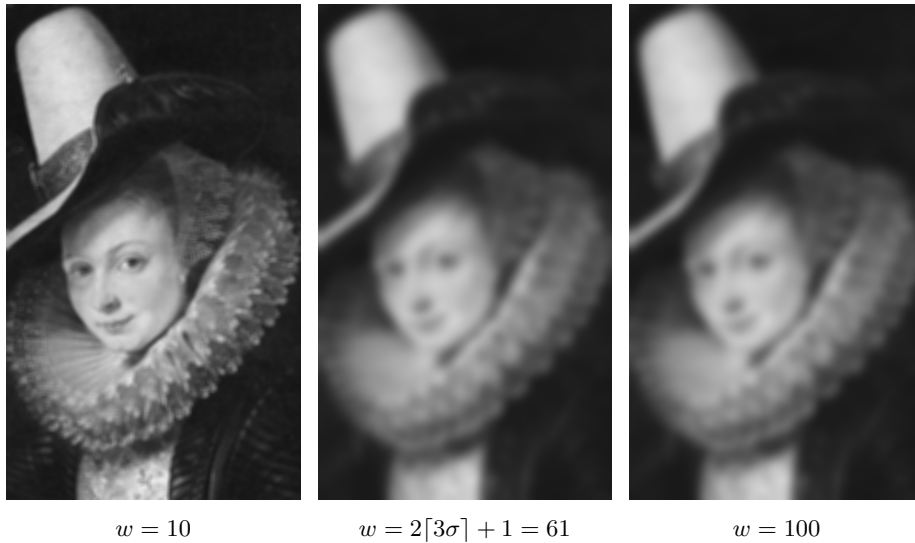


E1



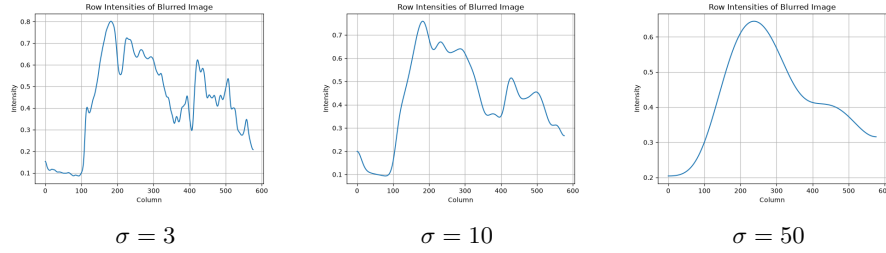
Effect of increasing σ on Gaussian blurring.

Increasing σ results in stronger blurring. That is because the Gaussian curve is stretched by σ ; therefore, more neighboring pixels are included in the mean calculation, and the middle pixel is weighted with decreasing amplitude.



Comparison of different kernel widths for Gaussian blurring with $\sigma = 10$.

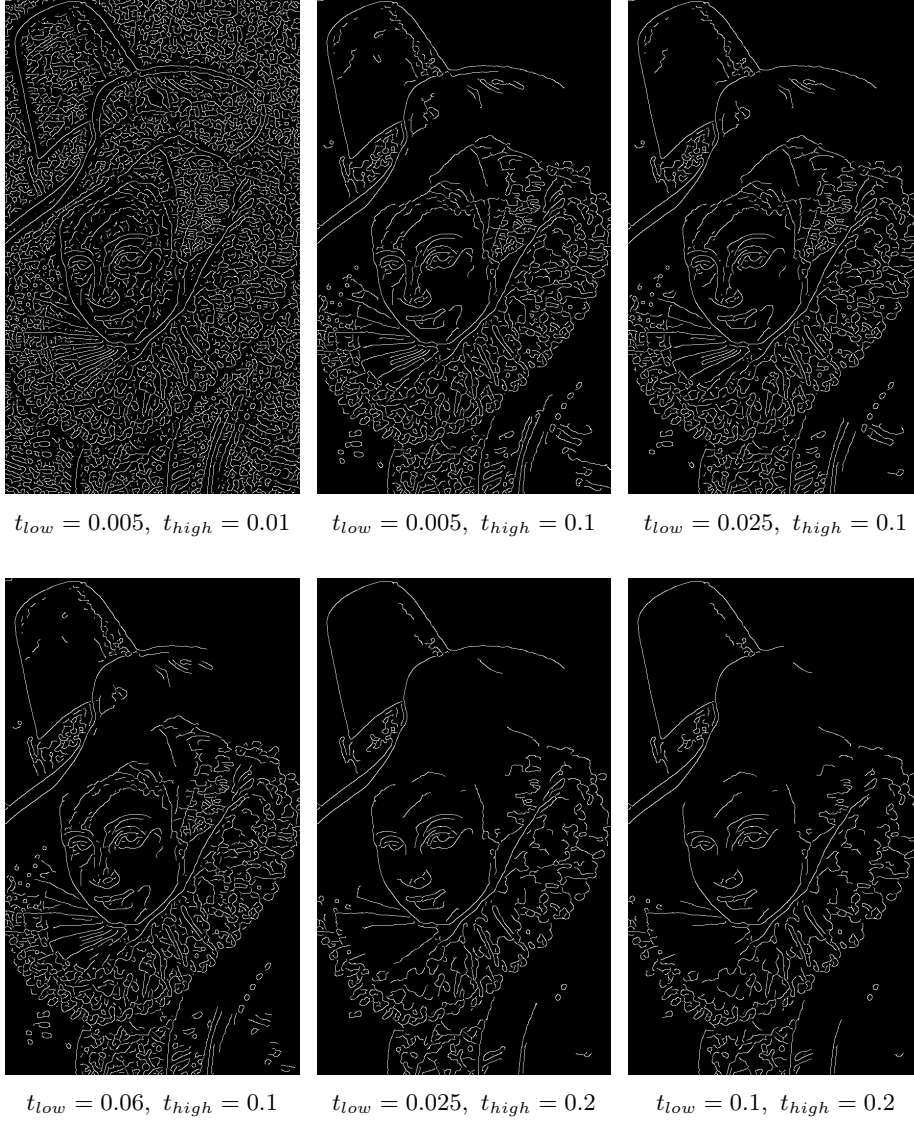
The kernel width is chosen as $k_w = 2\lceil 3\sigma \rceil + 1$ so that the kernel covers approx. 3σ around its center, where all relevant Gaussian weights are located. When the width is selected smaller than that, the image gets less blurred because the curve is cut off early, fewer neighbors are taken into account, and the middle pixels are weighted more strongly relative to the neighboring pixels. Selecting a wider width results in almost no visible change because the Gaussian weights outside the 3σ area are negligibly small.



Effect of Gaussian blurring on the pixel intensity profile.

Gaussian blurring smooths the intensity curve. Peaks and valleys are less pronounced and are spread over several pixels. This happens because pixels are calculated as a weighted average of themselves and their neighbors. With increasing σ , more and more pixels are taken into account, and therefore the curve gets flattened.

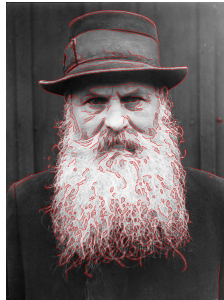
E2



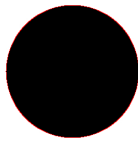
Effect of different hysteresis threshold values on the detected edges.

Increasing the hysteresis thresholds reduces the number of detected edges. A low t_{low} allows weak pixels to be considered, which results in more detail but also possibly cluttered images. Increasing t_{low} removes weak edge pixels and can cause weaker edge segments to disappear or become interrupted. Increasing t_{high} makes it harder for an edge component to contain a sufficiently

strong edge pixel. As a result, fewer connected edge components are kept. Very low values for both thresholds produce many weak and possibly irrelevant edges, while very high values remove weaker but still meaningful edges.



Beardman



Circle



Parliament



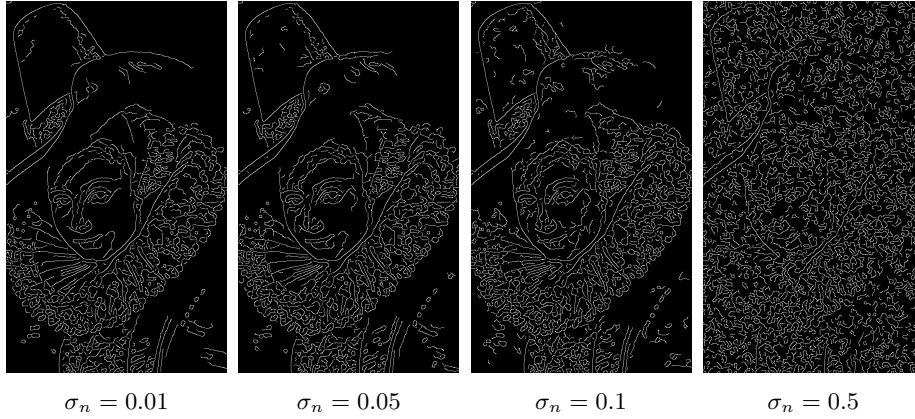
Rubens

Edge detection on different images using the same hysteresis thresholds $t_{low} = 0.1$ and $t_{high} = 0.2$.

Same thresholds can produce different results because the images contain different intensity distributions, contrasts, and amounts of detail, which result in different gradient strengths. Therefore, the same threshold values can remove relevant edges in one image while keeping many details in another image.

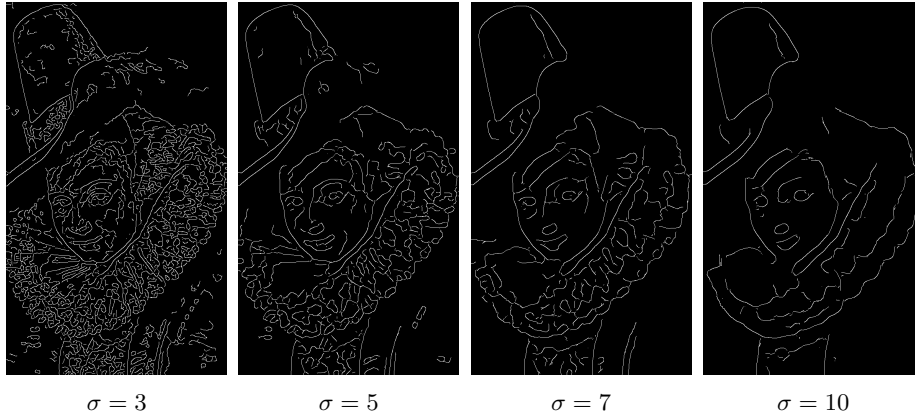
Keeping the thresholds the same can still be useful when processing images captured under similar conditions (lighting, contrast, etc.). One example of this could be quality control in an industrial setting, where produced materials are checked using images from one camera under stable conditions.

E3



Effect of increasing Gaussian noise on the detected edges.

Increasing the standard deviation of Gaussian noise increases the number of detected edges. This happens because the noise creates additional local intensity changes, which can produce gradients high enough to be detected as edges. As the noise level is increased, local noise gradient spikes are detected as edges, and at a standard deviation of 0.5, the real edges are hardly distinguishable from the noise-induced edges.



Effect of increasing Gaussian blur on an image with Gaussian noise $\sigma_n = 0.1$.

For higher σ values of the Gaussian blur, Gaussian noise gets filtered. Small local intensity fluctuations caused by the noise are smoothed out before the gradient is calculated. Therefore, fewer noise-induced edges are detected. With

a Gaussian blur of $\sigma = 10$, it is visible that almost no noise is left, but on the other hand, a significant amount of the weak "normal" edges is removed as well. This is to be expected because the blurring smooths gradients over the whole image, and first and foremost weak edges get lost. This shows that a compromise between noise reduction and preservation of relevant edges is necessary.